

Basic Cisco ASA 5506-x Configuration Example

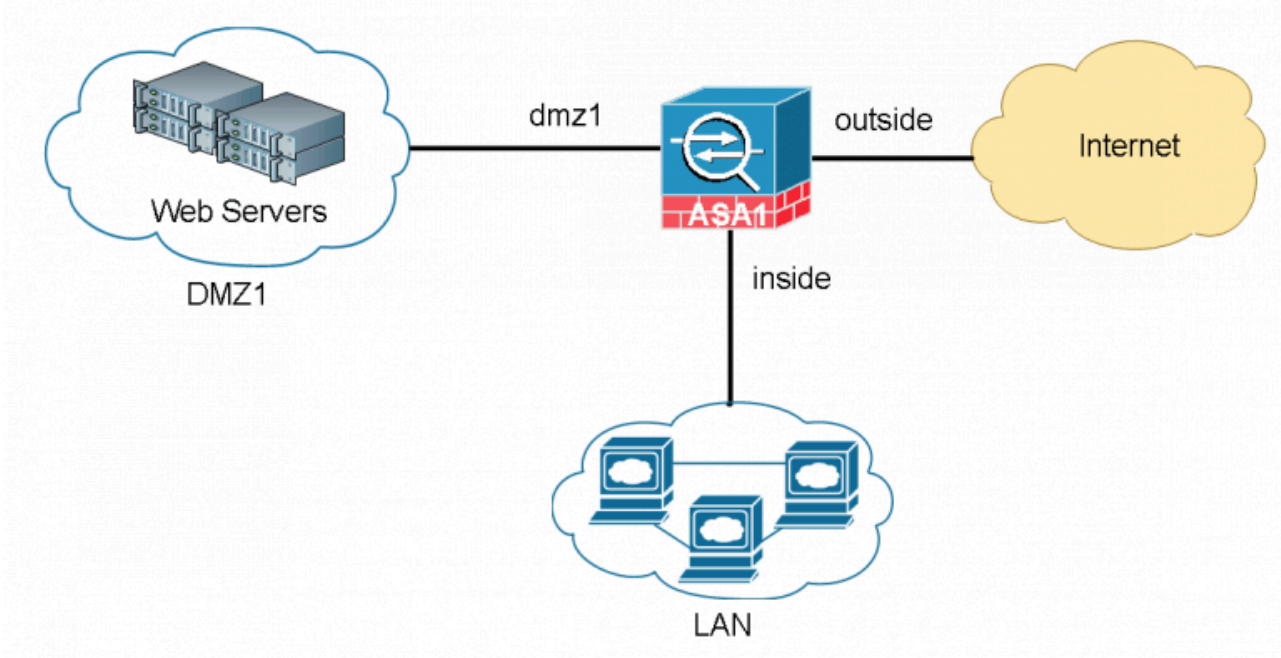
Cisco’s latest additions to their “next-generation” firewall family are the ASA 5506-X, 5508-X, 5516-X and 5585-X with FirePOWER modules. The new “X” product line incorporated the industry leading IPS technologies, provides next-generation Intrusion Prevention (NGIPS), Application Visibility and Control (AVC), Advanced Malware Protection (AMP) and URL Filtering. In the basic Cisco ASA 5506-x Configuration example, we will cover the fundamentals to setup an ASA firewall for a typical business network. FirePOWER module configuration is covered in a separate document. For a more comprehensive, multi-DMZ network configuration example please sees: [Cisco ASA 5506-X FirePOWER Module Configuration Example Part 1-4](#).

Below is the network topology that this example is based on. We will cover how to configure basic ACL (Access Control List), Network Address Translation (NAT) and a simple DMZ network hosting WWW server. The equipment used in this example is Cisco ASA 5506-X with FirePOWER module, running code 9.5(2).

You can download the entire lab setup and configuration files for FREE.

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Basic Cisco ASA 5506-x Configuration Example

Network Requirements

In a typical business environment, the network is comprised of three segments – Internet, user LAN and optionally a DMZ network. The DMZ network is used to host publically accessible servers such as web server, Email server and so on. The Cisco ASA acts as a Firewall, as well as an Internet gateway.

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- LAN users and Web Servers all have Internet access.
- LAN users have full access to the Web Server network segment (DMZ1) but DMZ1 does not have any access to the LAN (in case DMZ is compromised).
- Anyone on the Internet can access the Web Server via a publically NAT’d IP address over HTTP.
- All other traffic is denied unless explicitly allowed.

Update ASA software and ASDM code

Download the recent stable release from Cisco.com and transfer the codes to the ASA.

```
ASA1# dir
Directory of disk0:/

94      -rwx   69454656    06:49:06 Jul 13 2015  asa941-lfbff-k8.SPA
95      -rwx    33         12:21:36 Aug 12 2015  .boot_string
11      drwx   4096      06:51:46 Jul 13 2015    log
19      drwx   4096      06:52:52 Jul 13 2015    crypto_archive
20      drwx   4096      06:53:10 Jul 13 2015    coredumpinfo
96      -rwx   41836544    06:57:56 Jul 13 2015    asasfr-5500x-boot-5.4.1-211.img
97      -rwx   532115     05:43:22 Oct 01 2015    crash.txt
99      -rwx   82335584    07:51:02 Jan 18 2016    asa952-lfbff-k8.SPA
100     -rwx   25627616    07:52:40 Jan 18 2016    asdm-752.bin

7859437568 bytes total (4416327680 bytes free)
ASA1#
```

Set the system to boot to the new image. Configure the ASDM image to be used.

```
ASA1(config)# boot system disk0:/asa952-lfbff-k8.SPA
ASA1(config)# asdm image disk0:/asdm-752.bin
```

Write memory and verify the bootvar is set correctly. Reboot the system to load the new image.

```
ASA1# show bootvar

BOOT variable = disk0:/asa952-lfbff-k8.SPA
Current BOOT variable = disk0:/asa952-lfbff-k8.SPA
CONFIG_FILE variable =
Current CONFIG_FILE variable =
```

Security levels on Cisco ASA Firewall

Before jumping into the configuration, I’d like to briefly touch on how Cisco ASAs work in a multi-level security design. The concept is not Cisco specific. It applies to any other business grade firewalls.

By default, traffic passing from a lower to higher security level is denied. This can be overridden by an ACL applied to that lower security interface. Also the ASA, by default, will allow traffic from higher to lower security interfaces. This behavior can also be overridden with an ACL. The security levels are defined by numeric numbers between 0 and 100. 0 is often placed on the untrusted network such as Internet. And 100 is the most secured network. In our example we assign security levels as following: LAN = 100, DMZ1 = 50 and outside = 0.

LAN is considered the most secured network. It not only hosts internal user workstations as well as mission critical production servers. LAN users can reach other networks. However, no inbound access is allowed from any other networks unless explicitly allowed.

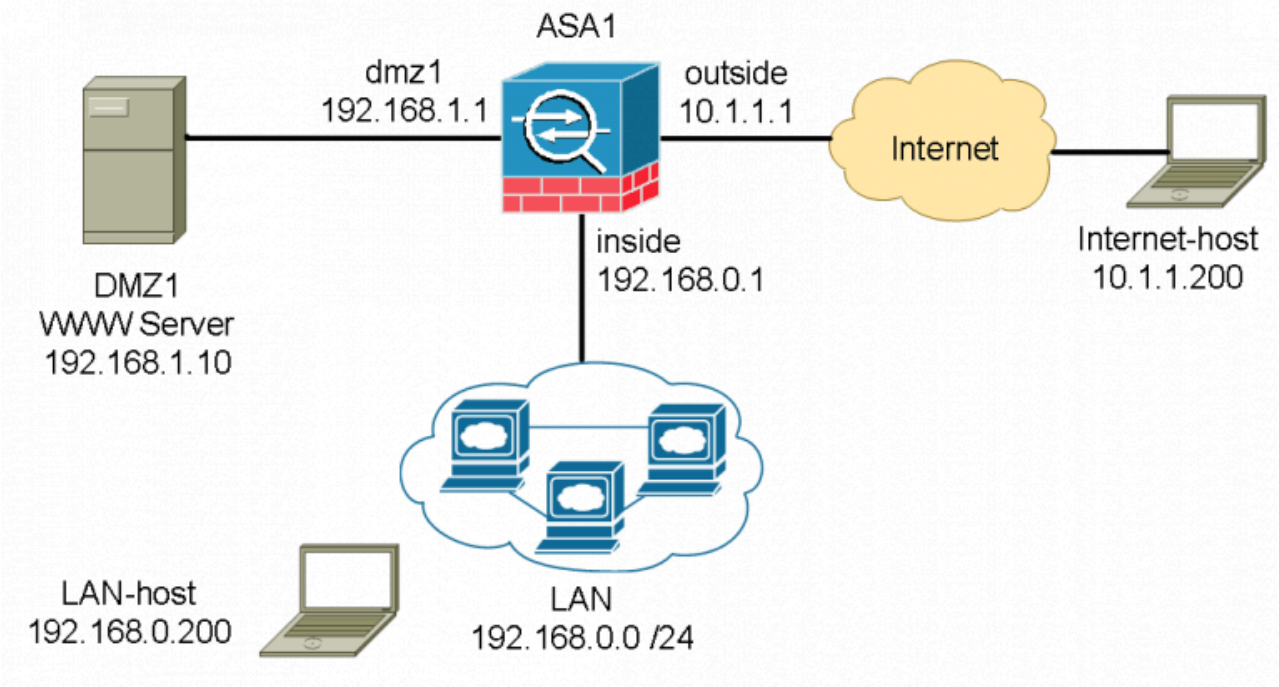
DMZ1 hosts public facing web servers. Any one on the Internet can reach the servers on TCP port 80 for HTTP.

The design idea here is that we don’t allow any possibilities of compromising the LAN. All “inbound” access to the LAN is denied unless the connection is initiated from the inside hosts. Servers in DMZ1 serve Internet web traffic and internal user traffic from the LAN.

ASA 5500-X	BGP	Cisco
Cisco Nexus	Cisco VIRL	
Conditional Routing		
Dynamips	Failover	
FirePOWER	GNS3	
import and export	Intel-VT	
Intrusion prevention systems		
IPS/IDS	IP SLA	Junip
kvm-ok	LWAP	NAT
NetApp	Pulse Secure	
SourceFire	SSL VPN	
topology	VMware	VPN
WAN	Wireless	WLC

Network Design and IP Assignment

For simplicity, we assume the SOHO network has less than 200 users and does not have a layer switch on the LAN. All user and server traffic point to the ASA as their default gateway to the Internet. We assign each network segment a /24 (255.255.255.0) subnet mask.



User LAN network:

Subnet: 192.168.0.0 /24
Gateway: 192.168.0.1 (ASA inside interface)
LAN-host (for testing): 192.168.0.200

DMZ1 network:

Subnet 192.168.1.0 /24
Gateway: 192.168.1.1
Web server: 192.168.1.10

Internet:

Internet-host (for testing): 10.1.1.200

Cisco ASA 5506-x Configuration

Step 1: Configure ASA interfaces and assign appropriate security levels

The ASA 5506-X comes with 8 GigE routed interfaces. We are going to use three of the interfaces in this network – inside (100), dmz1(50) and outside (0).

```
interface GigabitEthernet1/1
  description to WAN
  nameif outside
  security-level 0
  ip address 10.1.1.1 255.255.255.0
!
interface GigabitEthernet1/2
  description to LAN
  nameif inside
  security-level 100
  ip address 192.168.0.1 255.255.255.0
!
interface GigabitEthernet1/3
  description to DMZ1
  nameif dmz1
  security-level 50
  ip address 192.168.1.1 255.255.255.0
```

Step 2: Configure ASA as an Internet gateway, enable Internet access

There are two things required in order for the internal hosts to go out to the Internet, configuring Network Address Translation (NAT) and routing all traffic to the ISP. You do not need an ACL because all outbound traffic is traversing from higher security level (inside and dmz1) to lower security level (outside).

```
nat (inside,outside) after-auto source dynamic any interface
nat (dmz1,outside) after-auto source dynamic any interface
```

The configuration above states that any traffic coming from inside and dmz1 network, translate the source IP to the outside interface’s IP for outbound Internet traffic. The “after-auto” keyword simply set this NAT the least preferred rule to be evaluated after Manual NAT and Auto NAT are evaluated. The reason we want to give it the least preference is to avoid possible conflict with other NAT rules.

Next is configuring a default gateway and route all traffic to the upstream ISP. 10.1.1.2 is the gateway the ISP provided.

```
route outside 0.0.0.0 0.0.0.0 10.1.1.2
```

Also make sure “inspect icmp” is configured under global_policy. It allows icmp return traffic to pass the ASA while the Ping is initiated from inside hosts.

```
policy-map global_policy
class inspection_default
inspect icmp
```

At this point, you should be able to ping the host 10.1.1.200 on the Internet from any internal subnets.

Step 3: Configure static NAT to web servers, grant Internet inbound access to web servers

First we define two objects for the web server, one for its internal IP and one for its public facing IP.

```
object network WWW-EXT
  host 10.1.1.10
!
object network WWW-INT
  host 192.168.1.10
!
nat (dmz1,outside) source static WWW-INT WWW-EXT
```

Anyone on the Internet trying to access the web server, they’ll use the public IP defined in WWW-EXT. It will be translated to the private IP defined in WWW-INT.

Now the IP address translation has been done. We will need to configure ACL and allow Internet inbound traffic to access the web server. And apply the ACL to the outside interface.

```
access-list OUTSIDE extended permit tcp any object WWW-INT eq www
access-list OUTSIDE extended permit icmp any4 any4 echo
access-group OUTSIDE in interface outside
```

The ACL states, permit traffic from anywhere to the web server (WWW-INT: 192.168.1.10) on port 80. For troubleshooting and demonstration purpose, we also allow ICMP ping traffic. In a real-world network, I recommend disallow Ping for higher security.

Step 4: Configure DHCP service on the ASA

This step is optional. If you have a DHCP server on the LAN you can skip to the next step. For small businesses that do not have server in house, you may configure the ASA to be a DHCP server.

Specify a DHCP address pool and the interface for the client to connect. We reserve a few address before and after the pool for future network devices or appliances that require static IP.

```
dhcpd address 192.168.0.5-192.168.0.250 inside
```

Specify the IP address of the DNS servers for client use. It is always a good idea to have the secondary DNS server in case the primary fails.

```
dhcpd dns 9.9.9.9 4.2.2.2
```

Specify the lease length to be granted to the client. This lease equals the amount of time (in seconds) the client can use its allocated IP address before the lease expires. Enter a value between 0 to 1,048,575.The default value is 3600 seconds.

```
dhcpd lease 3600
dhcpd ping_timeout 50
```

Enable the DHCP service to listen for DHCP client requests on the enabled interface.

```
dhcpd enable inside
dhcprelay timeout 60
```

(Optional) Step 5: Redirect traffic to the FirePOWER module for deeper level inspection

In order to utilize any of the ASA’s next-generation firewall features, Cisco made customers order subscription based licenses for the FirePOWER module to work. The subscription based licenses can be purchased annually, 3 or 5 years with discount. Here are list of licenses available:

- Intrusion detection and prevention (IPS license)
- Application Visibility and Control (AVC)
- File control and advanced malware protection (AMP)
- Application, user, and URL control (URL Filtering)
- IPS license is required for the AVC, AMP and URL Filtering license.

If you have a FirePOWER feature license available and send traffic to the FirePOWER module for deeper level inspection, here is an example of send all traffic to FirePOWER. In case there was a software (in case of 5585-X, it is hardware) failure, bypass the FirePOWER module without inspection.

```
class-map global-class
  match any
policy-map global_policy
  class global-class
    sfr fail-open
```

Step 6: Hardening the device

Shutdown unused interfaces

```
interface GigabitEthernet1/4 through 1/8
shutdown
```

Enable SSH access for admin

There are three steps to enable SSH access:

1. Create a hostname for your ASA
2. Generate a RSA key
3. Configure SSH access to the ASA, and only allow from known IP/networks.

Configuration example:

```
ASA1(config)# hostname ASA1
ASA1(config)# crypto key generate rsa modulus 1024
WARNING: You have a RSA keypair already defined named <Default-RSA-Key
Do you really want to replace them? [yes/no]: yes
Keypair generation process begin. Please wait...
```

! The IP subnets from where you trust to manage the ASA

```
ssh 12.2.1.0 255.255.255.0 outside
ssh 192.168.0.0 255.255.0.0 inside
ssh timeout 30
ssh version 2
aaa authentication ssh console LOCAL
```

Step 7: Configure time and enable logging

It is important to enable logging so we know what happened in case there was an incident. Make sure time is set correctly and timestamp is enabled while logging. In this example we enabled logging into the ASA’s buffer memory. The maximum log size can grow up to 512MB and then the oldest logs are overwritten. The logging level is set to “debugging”, which records everything in detailed level.

```
ASA1# clock set 12:05:00 Jan 22 2016
ASA1# clock timezone EST -5
ASA1# clock summer-time EST recurring
ASA1# logging enable
ASA1# logging timestamp
ASA1# logging buffer-size 512000
ASA1# logging buffered debugging
```

To view logs, issue command “show logging” on the ASA.

For a more comprehensive, multi-DMZ network configuration example please read:

[Cisco ASA DMZ Configuration Example](#)

[Cisco ASA FirePOWER Module Configuration Example Part 1- 4](#)

You can download the entire lab setup and configuration files for FREE. The package includes:



- [INSTRUCTIONS.pdf](#)– Read this instruction first It covers what each downloaded file is for and how to use them.

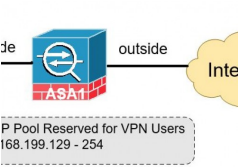
- Basic Cisco ASA 5506-x Configuration Example.pdf – The article in PDF format for your offline reference.
- Cisco-ASA5506-config.txt – The final configuration for the Cisco ASA. You may use it on any compatible ASA devices.
- Cisco_ASA5506-X.virl – Cisco VIRT topology file with final lab configuration. It is fully configured lab based on the requirements in the article.

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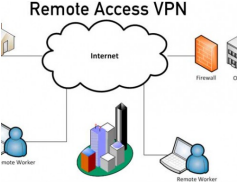
By [Jack Wang](#) | January 22nd, 2016 | [Security](#) | [14 Comments](#)

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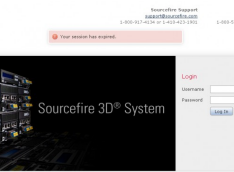
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Luke 8 years ago

Good article demonstrating the little amount of configuration required to get an ASA configured for use, you may want to include in the SSH section ‘aaa authentication ssh console LOCAL’ as I believe that is a requirement for SSH to authenticate after being enabled.

+ 0 - Reply



Jack Wang 8 years ago

Reply to Luke

Author

Good catch! I will make a note to add. Thank you.

+ 0 - Reply



Harry 8 years ago

Hello Jack, good article. By default, firepower don't have to be enabled or configured, right? I have question about asa 5506x, currently we bought and tested it. We have Verizon fios for business with 29 static IP addresses, Verizon tech instruct me to use subnet mask of 255.255.255.0. and that's working fine once I configure a labtop with any static IP Verizon provided. However, once I connect my asa to the Verizon network, it seems only one IP is working fine, that asa's outside IP address. I added static NAT one by one, it doesn't work. I believe it's Verizon up stream routers don't route traffic properly since once I turn on firewall log, there are no hits for all other IP addresses. I asked Verizon tech to route all public IP addresses they assigned to me to our asa's outside interface IP, they told me that we cannot do static route for us. They blame this is my asa configuration issue. Is there any workaround for this?

Many thanks.

Harry

+ 0 — ➔ Reply



Jack Wang 8 years ago

🔗 Reply to [Harry](#)

Author

Harry, there is a simple test you can do to isolate the problem. Use the same laptop you used for testing, assign the static IPs that you are supposed to have, one by one to test. For example, Verizon provided you 123.123.123.65 – 79, you should be able to use any of these IPs in that range on your laptop to get to Internet. If that test fails, it is Verizon's problem. If all IPs worked fine on your laptop, we'll have to look into your ASA configuration. Hope it helps.

+ 0 — ➔ Reply



Harry 8 years ago

🔗 Reply to [Jack Wang](#)

Hi Jack,

Thank you for your quick reply. Yes, I did one by one IP address test, it works fine. I enable proxy arp for NAT as well. It still doesn't help.

Thanks,
Harry

+ 0 — ➔ Reply



Jack WAng 8 years ago

🔗 Reply to [Harry](#)

Author

I'll need to see your ASA configuration to tell what's wrong.

+ 0 — ➔ Reply



Harry 8 years ago

🔗 Reply to [Jack WAng](#)

Hi Jack,
I have sent email to you.
Thank you.
Harry

+ 0 — ➔ Reply



Bil Herd 7 years ago

🔗 Reply to [Harry](#)

Hi Harry,
This is a known problem in this region, Verizon's equipment does an "illegal" ARP request from 0.0.0.0 which the ASA throws out.

A little more is at:
<http://bilherd.mercury-cg.com/?p=84>

+ 0 — ➔ Reply



Juergen 6 years ago

Shouldn't we better use object nat rules for nat from inside and dmz to the internet? With your suggested manual nat rules, we have to add a nat exemption for traffic between inside and dmz to make connections from inside to dmz and/or from dmz to outside possible. An object nat rule for the inside network from inside to outside would be restricted to the traffic that will be send to outside according to the current routing table, the twice nat rule will force all matching traffic (independent from the routing table) to leave the asa through the destination interface given in the nat rule, i.e. in this case the outside interface ... For identity nat rules, there is a "route-lookup" option, but this is only possible for identity nat ...

+ 0 - Reply



Jack 6 years ago

Reply to Juergen

Author

Juergen, your configuration would work but I found it is much cleaner to configure manual NAT as described in this article. When you have a lot of NAT'ing rules, the configuration can get large and difficult to read and manage. In addition, manual NAT is much more flexible and you can apply the same concept to all NAT'ing situations. Hope it helps.

+ 0 - Reply



Ryan 6 years ago

Nice article, Jack. One thing I think you might have missed is the physical cabling and interface setup of the firepower module... at least on a 5506.

+ 0 - Reply



Jack 6 years ago

Reply to Ryan

Author

Ryan, thanks for your suggestions. I'll definitely incorporate physical cabling and interface setup in future articles.

+ 0 - Reply



Horst 5 years ago

Good artickel! I have one question:
If i want the description to WAN to get the ip by dhcp from the Provider
witch route outside 0.0.0.0 0.0.0.0 10.1.1.2 is to use?

Thank you!

+ 0 - Reply



Jack Wang 5 years ago

Reply to Horst

Author

Horst, in that case please use the configuration below. The ASA will automatically obtain an IP from your ISP and set default gateway to the ISP. No need to configure static route.
ASA#configure terminal
ASA(config)#interface GigabitEthernet1/1
ASA(config-if)#ip address dhcp setroute

+ 0 - Reply

